The dynamics of MM: Einstein's weak field, Friedmann's cosmology, and a boundary set by the Galaxy, with a registered test on Gaia DR4 wide binaries

Ugo Lanciano
Independent researcher, Locri, Italy · ORCID 0009-0008-7618-9635

Version 2.70, 2026-10-28. Archived on Zenodo, https://doi.org/10.5281/zenodo.23730218. Licence CC BY 4.0. Developed with the help of AI assistants, as declared in the record.

Abstract

This record gives the dynamics of Mathematical Materiality (MM). In the weak field it is linearised Einstein gravity, and the homogeneous model is Friedmann's. The junction to the $a_0 = cH_0/6$ layer is read as a boundary set by the Galaxy (the rule L*), and a premise of the author gives that boundary with two declared choices. The record also registers, before Gaia DR4 (planned for 2 December 2026), a test of this reading on wide binaries, with rules fixed before the data. It is one of MM's checks: the outcome concerns one reading of MM, not the framework. The dynamics: The prediction. In MM's reading L*, a pair of stars reads only its own field inside a boundary, where its own Newtonian field exceeds the Galactic one (7 773 AU for 1.5 solar masses near the Sun). The MOND-like boost is then the same in every direction. QUMOND with the external field predicts instead, inside that boundary, that pairs aligned with the Galactic field move more slowly. Newton predicts no difference.

Version 2.70 of the Mathematical Materiality (MM) series, October 2026. The dynamics of MM, and a registration record deposited before Gaia DR4 (planned for 2 December 2026).

- **Part A** is a prediction with its rule, simulations and a first look at DR3.
- **Part B** is the dynamics of MM from which the prediction comes: linearised Einstein gravity in the weak field, Friedmann's equations for the homogeneous model, and a boundary set by the Galaxy for the junction to the a_0 layer.
- **Part C** is a premise of the author that gives the rule behind the prediction, and a first look at what it predicts for the amplitude, with a registered rule for DR4 and an estimate of its power.

The working documents are in Italian; this README summarises them. The record continues MM 2.56 (DOI 10.5281/zenodo.23506401), the registered amplitude test of wide binaries for DR4, and reuses its download code.

This is one of MM's checks. The outcome concerns one reading of MM (called L* below), not the framework.

Part A — the prediction

A1. The question

Near the Sun, a wide binary sits in the gravitational field of the Galaxy. Two readings make different predictions for the pair's internal motion.

reading	inside the boundary radius R_b	outside
L* (MM)	the pair reads only its own field: a MOND-like boost from its own field, **the same in every direction**	QUMOND with the Galactic external field
QUMOND + external field (MOND)	the Galactic field enters everywhere: the boost is **weaker along the direction of the Galactic field**	the same
Newton	no boost, no dependence on direction	the same

- **Boundary.** R_b is where the pair's own Newtonian field equals the Galactic one. For 1.5 M☉ near the Sun it is R_b = 7 773 AU, that is 0.861 r_M, with $r_M = \sqrt{(GM/a_0)}$ and $a_0 = cH_0/6 = 1.0914 \cdot 10^{-10}$ m s^{-2}.

- **Field.** The Galactic Newtonian field near the Sun is e_N = 1.3488 a_0 (V_0 = 233 km/s, R_0 = 8.21 kpc).
- **Observable.** Inside the boundary, pairs whose projected separation is aligned with the projected direction of the Galactic centre should move as fast as pairs seen across it (L*, Newton), or more slowly (QUMOND).
- **Coverage.** The test separates QUMOND from {L*, Newton}. The amplitude test of MM 2.56 separates Newton from the others.

A2. Observable, data and rule (fixed before the data)

- **Sample.**
- The wide binaries of El-Badry, Rix & Heintz (2021), already downloaded on 16 October 2026 for MM 2.56.
- The quality cuts frozen on 11 October 2026: R < 0.01, RUWE < 1.4, MSMS, d < 200 pc, ϖ/σ_ϖ > 20, consistent parallaxes, 3 < M_G < 12, proper-motion error on u < 0.1, depth term < 0.1, u < 3.
- Photometric masses; x_p = s/r_M with the projected separation s.
- **Velocity.** u = |v_⊥|/v_N, the component of the relative sky velocity perpendicular to the projected separation (no perspective term), with v_N = √(GM/s).
- **Angle.** φ is the angle between the projected separation and the projected direction of the Galactic centre.
- Classes: aligned if |cos φ| ≥ 1/√2, across otherwise.
- Pairs within about 17° of the centre or anticentre on the sky are removed (|ê_p| < 0.3).
- **Statistic.** Δ_φ = ⟨u⟩_aligned / ⟨u⟩_across − 1, in five bands of x_p. The key band is 0.3-0.86, inside the boundary.
- **Uncertainty.** Bootstrap over pairs, 2 000 resamples. No star appears in more than one pair.
- **Reading rule (key band).**
- **«No QUMOND anisotropy inside the boundary»** (favours L* or Newton) if the data lie more than 2σ beyond the QUMOND interval on the side of L\, *and within 2σ of the L* interval.
- **«QUMOND-like anisotropy»** in the opposite case.
- **«Not decidable»** otherwise.
- **Control.** In the control band (0.05-0.2) all readings predict zero to 2 %. If the data there exceed 3σ, the reading carries «systematic in the control».

A3. Predictions (simulated before opening the data)

- **Models.** Newton, QUMOND+EFE with the vector force (radial and transverse components, from a three-dimensional integral of the phantom density for two equal masses), and L* with a smooth, energy-conserving junction at the boundary.
- **Population.** 4 000 orbits, isotropic, apocentres log-uniform in 0.15-5 r_M, p(e) ∝ e^1.2. Integrated for 40 periods with an adaptive leapfrog. Seen from 4 random lines of sight.
- **Measurement.** The same as the data: mass and distance errors, noise on u, toy contaminants, the same cuts.
- **Runs.** Computed on AWS before the data were opened.

band (x_p)	Newton	QUMOND+EFE	L*	expected σ with DR3
control (0.05-0.2)	[−0.006; +0.017]	[−0.006; +0.020]	[−0.005; +0.019]	0.011
inner (0.2-0.3)	[−0.007; +0.022]	[−0.014; +0.022]	[−0.007; +0.023]	0.025
key (0.3-0.86)	[−0.015; +0.007]	**[−0.031; −0.004]**	**[−0.012; +0.012]**	0.022
outer (0.86-1.5)	[−0.022; +0.006]	[+0.011; +0.035]	[+0.007; +0.030]	0.048
far (1.5-3.3)	[−0.012; +0.018]	[+0.019; +0.048]	[+0.020; +0.046]	0.082

- **Intervals** span two time windows (periods 0-10 and 30-40), each ± its simulation uncertainty.
- **Built-in check:** Newton gives zero, as it must.
- **Paired difference** L* − QUMOND in the key band: +1.1 % ± 0.4 % (periods 0-10) and +2.5 % ± 0.8 % (30-40).
- **Expected power with DR3: 0.8σ.** «Not decidable» was written down as the most likely outcome before the data.

A4. First look at DR3

- **Key band:** 4 140 pairs, Δ_φ = **+0.018 ± 0.027**. That is 0.2σ from the L* interval, 0.4σ from Newton and 0.8σ from QUMOND. **Reading: not decidable with DR3.**

- **Control band:** 17 128 pairs, +0.027 ± 0.013 (2.1σ, below the 3σ flag).
- The control band of an earlier test (F-46, August 2026, 300-2 000 AU) also showed a small excess of the same sign.
- The two are not fully independent. A possible orientation-dependent systematic has to be understood before DR4.
- **Spread of u:** wider in the data than in the simulations (coefficient of variation 0.84-0.87 against 0.69-0.70), so the real uncertainty is larger than expected.

A5. Rule for DR4 (exact replica)

- **Same rule.** The same pairs, re-measured in DR4, go through the same band, statistic, predictions and rule.
- **Pipeline:**
- download: `scarica_dr4.py` of MM 2.56, mode `dr4`, fallback `dr4_posizione`;
- `code/dr4/converti_p10.py`: Gaia archive format → catalogue format; pairs with a DR4 non-single-star solution are removed;
- `costruisci_p10_dr4.py`;
- `analisi_p10_dr4.py`.
- **Validity:** at least 3 000 pairs in the key band and 1 000 per class.
- **Dry run on DR3:**
- the copied sample builder reproduces the DR3 sample bit for bit;
- 3 000 pairs downloaded from the Gaia archive in October reproduce the same pairs, bands and classes, with u within 0.27 %.
- **Expected outcome.** About ten times the DR3 pairs would be needed to reach 2σ with this observable, so «not decidable» remains the most likely outcome with DR4. The record fixes the test in time.
- **Not in this record:** a more powerful analysis (shape of the distribution, all bands, per-pair errors). If done, it will be registered separately before 2 December.

A6. What the prediction rests on

- **The reading L*.** A pair is read as isolated inside a sharp boundary. In MM this is a rule, not a derived law (Part B). Part C shows that a premise of the author, with two declared choices, gives the boundary and the isotropy inside it.
- **The junction at the boundary.**
- With a sharp switch (isotropic inside, anisotropic outside) the force does not conserve energy.
- The smooth, conservative junction used here, a shell from 7 773 to 8 550 AU, is the author's choice. The sensitivity to other smooth junctions has not been measured.
- **An idealised population.**
- Known starting distribution, fixed direction of the external field, equal masses in the dynamics, no tides or encounters.
- In real data the population has to be estimated, and part of the signal may be absorbed. The simulated numbers are lower bounds on the number of pairs needed.

Part B — the dynamics of MM behind the prediction

MM reads gravity from counts on a graph, with the closure ψ + Nψ = 2. Since 26 October 2026 the dynamics was worked out level by level, and each step was registered before computing (fingerprints and UTC times in `registers/`). The notes are in `B_dynamics/notes/`, in Italian.

B1. What holds

- **Level 0** (inventory): the constraints of the dynamics are already in MM's Regge-type action.
- **Level 1:**
- the linearised Bianchi identity follows from the lattice action (with a binding stress);
- read as a law in motion, the closure adds a scalar mode if the trace varies.
- **Weak field:** linearised Einstein gravity.
- **Level 3** (cosmology): the local homogeneous model is exactly Friedmann. The density ρ(a) is not fixed by the framework.

B2. Matter in motion (Level 2)

- **Compact stars.** A distant reading held fixed (D1-0) does not admit generic motions. A «breathing» distant reading (D1-a) gives:
- a main mode at 6.68 kHz (SLy) and 7.14 kHz (APR4);
- tidal deformability +22 % compared with general relativity.
- **Open:** the stability of D1-a is not decided.

B3. Clusters (Level 3)

- Without a_0, the dynamics of Levels 1-2 is Newtonian: a factor 6.7 of mass is missing.
- With a_0 a factor 1.5-2.2 remains, as in MOND.

B4. The junction between the closure and the a_0 layer (the open node)

step	what was tried	outcome
1	algebraic hook $L = 4(\psi_0 - 1)$	not a derived junction; K1 fails on KiDS, the QUMOND realisation fails on Cassini
2	rule L*: inside R_b each system reads its own field	passes Cassini by construction; KiDS allowed; binaries plateau
3	local dynamics: QUMOND, a ceiling, a curvature reading	QUMOND 18.5σ and ceiling 13.7σ on Cassini (Saturn Q_2); curvature fails granularity
4	AQUAL; a field-curvature combination (D3)	AQUAL 23.5σ on Cassini; D3 passes Cassini, fails in star-poor discs
5	an extra field (AeST type), cubic Galileon	not decidable at the threshold; tidal quadrupole grows inward
6	L* written as a non-local law, with a rule for systems	consistent trees; two-body force validated
7	two-body QUMOND + external field in 3D	+6 % step at the boundary relative to the isotropic mean
8	population with the mean (central) force	$N_3 = 1.3 \cdot 10^5$ pairs (3σ): «uncertain»
9	population with the vector force	$N_3 = 1\text{-}2.5 \cdot 10^4$ (idealised): the anisotropy is the main signature
10	DR3 test (Part A)	not decidable with DR3
11, 11b, 11c	the author's premise on the observation MCD (Part C)	gives the boundary and isotropy of L*; on DR3 its law fits best, a posteriori; DR4 is not expected to separate it from the chain

Step 9 in more detail.

- The transverse component of the external field is not small: at the boundary $U \approx -0.16$, against ν_e $K = -0.385$ in the linear limit.
- It makes orbits more eccentric over tens of periods (an analogy with Kozai cycles). Inside the boundary L* switches it off.
- Six registered corrections:
- non-commensurate sampling;
- a late time window;
- a conservative QUMOND field from a potential (V8: the computed transverse component agrees with the potential-derived one to 6 %);
- a smooth junction;
- an adaptive integrator;
- a convergence run with twice finer substeps and the energy checked over all 40 periods. It converges: $N_3 = 2.6 \cdot 10^4$ and $1.0 \cdot 10^4$, within −5 % and +1 % of the official run (NOTA_ESITO_PASSO9_v3.md).

B5. In one line

- **Where the dynamics coincides with Einstein,** it holds.
- **The part proper to MM** ($a_0 = cH_0/6$ and the junction to it) is still a rule, L*, not a derived law.
- **Local versions of the junction** tried so far fail on Cassini or on granularity.
- **L* makes a falsifiable prediction for wide binaries:** Part A.

Part C — a premise that gives L*, and what it predicts for the amplitude

C1. The premise and its translation (step 11)

- **The premise** (the author, 28 October 2026, in Italian in `C_premise/notes/PREMESSA_UGO_20261028.md`): the galaxy a system sits in acts on the system's observation MCD, which in turn determines the system's own field.
- **The translation is mine,** with two declared choices:
- the Galactic field enters as a scale (a number), not as a vector, so the reading is isotropic;
- the boost with which a system is read cannot exceed the boost with which the Galaxy reads the point where the system sits: ν_eff = min(ν(y), ν(e_N)), that is y_eff = max(y, e_N), with ν(y) = 1/(1 − e^(−√y)).

I call it T_max. Two other translations (a sum and a quadrature) give no sharp boundary. Other choices are possible, and «why the minimum» is not derived.

- **What follows:**
- **the boundary and the isotropy of L*:** *the boundary falls at y = e_N (0.861 r_M, 7 773 AU for 1.5 M☉), and inside it the pair reads only itself, the same in every direction. In L** both were posited;
- **outside the boundary T_max and L* part ways:** T_max gives an isotropic boost ν(e_N) = 1.455 and no step at the boundary; L* (QUMOND with the external field) gives 1.26-1.33, with the anisotropy;
- **Solar System:** anomalous acceleration of order 10^{-341} m s^{-2} at 9.5 AU. It passes;
- **NGC 2419:** ν ≈ 3.3-3.4 at 86 pc, against $\lesssim$ 1.1. It fails, as the book finds for every law of the internal and external accelerations alone. The book's way out is the cluster's history, which the premise does not replace.

C2. T_max against DR3, with the machinery of MM 2.56 (step 11b)

- **Registered before computing:**
- orbit libraries for T_max (200 000 orbits each, the integrator of MM 2.56, only the force changed);
- the forward simulations with hidden companions and the Poisson likelihood on 26 classes per band, as in MM 2.56;
- the same thresholds as MM 2.56 (5, and 2 for each subset with one replica removed);
- a sensitivity variant (2c) with the two-body reading of step 6 inside the boundary.
- **Checks.** The libraries pass the MM 2.56 criteria (energy error at the 99th percentile $9\cdot10^{-4}$, no unbound initial state). With its own six hypotheses the evaluation reproduces MM 2.56 unchanged (D = +7.64).
- **DR3 development data** (El-Badry pairs; 913 in the low-acceleration band, 4 000 simulated in the control band). Log-likelihood from the best hypothesis:

hypothesis	ΔlogL
T_max	**0**
T_max, two-body variant	−0.6
MOND, simple	−1.1
MOND, standard	−1.9
MM chain 674 (QUMOND, $a_0 = cH_0/6$; L* outside the boundary)	−3.4
MM chain 730	−5.3
MLS, $a_0 = 1.2\cdot10^{-10}$	−9.8

- **Outcome by the registered criterion:**
- T_max against the MM chain: +3.4 (subsets 0.2, 9.1, 5.3): **not decidable**;
- T_max against Newton: +11.1 (subsets 7.8, 13.6, 7.4): **T_max favoured**;
- the registered two-body variant (equal masses) gives the same classes. This does not show insensitivity to every two-body treatment.
- **Where the difference comes from.** Almost all of it is in the low-acceleration band, where T_max is an isotropic boost of 1.455.
- **A posteriori.** The idea of T_max outside the boundary came partly from these data: the best MOND hypothesis of MM 2.56 has a boost of about 1.45 there. On DR3 this is coherence, not a test. Differences of 1-2 among the MOND-like hypotheses are within the noise of about ±3 seen in MM 2.56.

C3. Rule for DR4

- **The same chain as MM 2.56** (download, conversion, forward simulations of its six hypotheses), then `esegui_p11b_dr4.sh`: `forward_tmax.py` for the two registered libraries and `valuta_tmax_c1.py` in mode `dr4`. It adds to the MM 2.56 rule without changing it.
- **Frozen with the rule:** 4 000 simulated pairs in the control band, as in MM 2.56; the thresholds 5 and 2 are an operational pre-registered criterion, not calibrated with false-positive simulations.
- **DR4 is not an independent replica of DR3:** the pairs are largely the same, measured again with more data. The test is prospective because the rule is fixed before the data.
- **Expected outcome: not decidable (step 11c, registered before computing).** Mock data were made with the same machinery: 40 mock experiments for each true hypothesis, with 1 to 8 times the DR3 pairs of the low-acceleration band.
- With the DR3 pairs (as in the DR4 replica) the rule picks the right reading with probability 0.12 (5/40) if T_max is true and 0.03 (1/40) if the chain is true: «with the same pairs DR4 probably does not separate them».
- With 8 times the pairs it is still only 0.38 and 0.05: the advantage does not grow. The 50 % and 80 % targets are not reached within 7 304 simulated pairs; beyond that range the threshold is not estimated.
- When the chain is true, the criterion favours T_max by mistake in 3 of 40 experiments at the DR3 size (0.07-0.12 at the various sizes), a very uncertain Monte Carlo estimate.
- **A compatible explanation (an a posteriori diagnostic, registered before it was run).** T_max and the chain differ mainly in the amplitude of the boost, and a change of amplitude can be absorbed by the mass scale, which the rule leaves free and common to both bands: with 8 times the pairs each reading moves towards the other by changing the masses by about 7 %.
- With the mass scale fixed exactly at the true value, 8 times the pairs give 0.82 and 0.65. This is a favourable case, not a requirement; the precision on the masses that would suffice is not simulated.
- The diagnostic does not isolate the cause: the Monte Carlo noise of the control-band models (fixed at the three registered replicas) and the coarse grids are not excluded.
- **The +3.4 seen on DR3 does not choose between them:** in a Gaussian approximation of the simulated distributions the density ratio is about 1.1 (descriptive, a posteriori).
- **Two checks of T_max with DR4:**
- the amplitude, by this rule;
- the direction, by Part A: no anisotropy inside the boundary, as for L\. *Outside the boundary T_max predicts none, while L* and QUMOND predict +0.7 % to +4.8 %; with this observable those bands are not decidable.

C4. What it means

- **The premise, with two declared choices, turns the rule L* into a consequence** inside the boundary, and it adds a prediction outside.
- **On DR3, a posteriori,** its law fits best of the eight hypotheses, but it is not distinguishable from the MM chain.
- **With DR4 and the registered rule,** the expected outcome is again «not decidable» (step 11c). The amplitude test separates both MOND-like readings from Newton; it hardly separates them from each other.
- **The outcomes concern my translation T_max,** not the premise nor MM.

Verification

- Every step was checked by two AI verifiers, «Astra» (Codex, GPT-6) and «Argon» (Gemini), in one or two rounds. Their reports and the author's answers are in the `verification/` folders.
- For step 9 the last round left one point open (the convergence of the official run), answered by correction 6, which converges.

- The author is not a professional physicist. Computations were done with AI assistants (Claude, checked by Codex and Gemini), not reviewed by a human referee.

Contents

- **A_prediction/documents:** protocol, predictions, outcome note and DR4 rule (Italian).
- **A_prediction/code:** sample builder, simulation and analysis; dr4/ holds the replica scripts.
- **A_prediction/data:** sample with Gaia identifiers; query and checksums of the El-Badry download.
- **A_prediction/results:** simulations (npz, json, logs), DR3 analysis, DR4 dry run.
- **B_dynamics/notes:** final notes of Levels 0-3 and of steps 1-9 of the junction, including the step-9 protocol and corrections.
- **B_dynamics/code** and **B_dynamics/results:** the codes and outputs of the binary chain (steps 6-9).
- **C_premise:** the premise, protocols, notes, code, outputs and verification of steps 11, 11b and 11c (the mock models and experiments of step 11c are in `results/p11c_modelli` and `results/p11c_pseudo`). The two orbit libraries (217 MB each) are not included; they are rebuilt by `esegui_p11b.sh` (seeds 17 and 18, about 10 minutes each), and their SHA-256 are in `registers/`.
- **registers:** REGISTRO.txt of each working folder, with the SHA-256 and UTC registration time of every file.
- **IMPRONTE.sha256:** checksums of all files in this record.

Paths. The programs use the author's paths. The El-Badry download (167 301 pairs, 138 MB) is not included; the query and checksums are.

References

Works cited in the record, with their DOI.

- A (1995). Thermodynamics of Spacetime: The Einstein Equation of State. https://doi.org/10.1103/physrevlett.75.1260
- A (2016). Entanglement Equilibrium and the Einstein Equation. https://doi.org/10.1103/physrevlett.116.201101
- Abbott et al. (2017). Gravitational Waves and Gamma-Rays from a Binary Neutron Star Merger: GW170817 and GRB 170817A. https://doi.org/10.3847/2041-8213/aa920c
- Abbott et al. (2017). GW170814: A Three-Detector Observation of Gravitational Waves from a Binary Black Hole Coalescence. https://doi.org/10.1103/physrevlett.119.141101
- Abbott et al. (2017). GW170817: Observation of Gravitational Waves from a Binary Neutron Star Inspiral. https://doi.org/10.1103/physrevlett.119.161101
- Abbott et al. (2017). Search for Post-merger Gravitational Waves from the Remnant of the Binary Neutron Star Merger GW170817. https://doi.org/10.3847/2041-8213/aa9a35
- Abbott et al. (2018). GW170817: Measurements of Neutron Star Radii and Equation of State. https://doi.org/10.1103/physrevlett.121.161101
- Abbott et al. (2019). Tests of General Relativity with GW170817. https://doi.org/10.1103/physrevlett.123.011102
- Abbott et al. (2019). Properties of the Binary Neutron Star Merger GW170817. https://doi.org/10.1103/physrevx.9.011001
- Abbott et al. (2020). GW190425: Observation of a Compact Binary Coalescence with Total Mass $\sim 3.4\ M_\odot$. https://doi.org/10.3847/2041-8213/ab75f5
- Abbott et al. (2021). Tests of general relativity with binary black holes from the second LIGO-Virgo gravitational-wave transient catalog. https://doi.org/10.1103/physrevd.103.122002
- Abbott et al. (2025). Tests of general relativity with GWTC-3. https://doi.org/10.1103/physrevd.112.084080
- Adhikari et al. (2021). Probing Galaxy Evolution in Massive Clusters Using ACT and DES: Splashback as a Cosmic Clock. https://doi.org/10.3847/1538-4357/ac0bbc
- Aker et al. (2022). Direct neutrino-mass measurement with sub-electronvolt sensitivity. https://doi.org/10.1038/s41567-021-01463-1
- Aker et al. (2025). Direct neutrino-mass measurement based on 259 days of KATRIN data. https://doi.org/10.1126/science.adq9592
- Andersson et al. (1998). Towards gravitational wave asteroseismology. https://doi.org/10.1046/j.1365-8711.1998.01840.x
- Angus et al. (2008). X-ray group and cluster mass profiles in MOND: unexplained mass on the group scale. https://doi.org/10.1111/j.1365-2966.2008.13353.x
- Arnowitt et al. (2008). Republication of: The dynamics of general relativity. https://doi.org/10.1007/s10714-008-0661-1
- Bertotti et al. (2003). A test of general relativity using radio links with the Cassini spacecraft. https://doi.org/10.1038/nature01997
- Biviano et al. (2013). CLASH-VLT: The mass, velocity-anisotropy, and pseudo-phase-space density profiles of

thez= 0.44 galaxy cluster MACS J1206.2-0847. https://doi.org/10.1051/0004-6361/201321955
- Biviano et al. (2023). CLASH-VLT: The Inner Slope of the MACS J1206.2-0847 Dark Matter Density Profile. https://doi.org/10.3847/1538-4357/acf832
- Cao et al. (2017). Space from Hilbert space: Recovering geometry from bulk entanglement. https://doi.org/10.1103/physrevd.95.024031
- Cao et al. (2018). Bulk entanglement gravity without a boundary: Towards finding Einstein's equation in Hilbert space. https://doi.org/10.1103/physrevd.97.086003
- Chang et al. (2018). The Splashback Feature around DES Galaxy Clusters: Galaxy Density and Weak Lensing Profiles. https://doi.org/10.3847/1538-4357/aad5e7
- Diaferio (1999). Mass estimation in the outer regions of galaxy clusters. https://doi.org/10.1046/j.1365-8711.1999.02864.x
- Eckert et al. (2018). Non-thermal pressure support in X-COP galaxy clusters. https://doi.org/10.1051/0004-6361/201833324
- Ettori et al. (2018). Hydrostatic mass profiles in X-COP galaxy clusters. https://doi.org/10.1051/0004-6361/201833323
- Everitt et al. (2011). Gravity Probe B: Final Results of a Space Experiment to Test General Relativity. https://doi.org/10.1103/physrevlett.106.221101
- Famaey et al. (2012). Modified Newtonian Dynamics (MOND): Observational Phenomenology and Relativistic Extensions. https://doi.org/10.12942/lrr-2012-10
- Flanagan et al. (2008). Constraining neutron-star tidal Love numbers with gravitational-wave detectors. https://doi.org/10.1103/physrevd.77.021502
- Gourgoulhon (2007). 3+1 Formalism and Bases of Numerical Relativity. https://doi.org/10.48550/arxiv.gr-qc/0703035
- Gunn et al. (1972). On the Infall of Matter Into Clusters of Galaxies and Some Effects on Their Evolution. https://doi.org/10.1086/151605
- Hinderer (2008). Tidal Love Numbers of Neutron Stars. https://doi.org/10.1086/533487
- Hinderer (2009). ERRATUM: "TIDAL LOVE NUMBERS OF NEUTRON STARS" (2008, ApJ, 677, 1216). https://doi.org/10.1088/0004-637x/697/1/964
- Ibata et al. (2011). THE GLOBULAR CLUSTER NGC 2419: A CRUCIBLE FOR THEORIES OF GRAVITY. https://doi.org/10.1088/0004-637x/738/2/186
- Israel et al. (2005). The Discovery of Rapid X-Ray Oscillations in the Tail of the SGR 1806-20 Hyperflare. https://doi.org/10.1086/432615
- Kramer et al. (2021). Strong-Field Gravity Tests with the Double Pulsar. https://doi.org/10.1103/physrevx.11.041050
- Merten et al. (2015). CLASH: THE CONCENTRATION-MASS RELATION OF GALAXY CLUSTERS. https://doi.org/10.1088/0004-637x/806/1/4
- More et al. (2015). THE SPLASHBACK RADIUS AS A PHYSICAL HALO BOUNDARY AND THE GROWTH OF HALO MASS. https://doi.org/10.1088/0004-637x/810/1/36
- More et al. (2016). DETECTION OF THE SPLASHBACK RADIUS AND HALO ASSEMBLY BIAS OF MASSIVE GALAXY CLUSTERS. https://doi.org/10.3847/0004-637x/825/1/39
- Oppenheimer et al. (1939). On Continued Gravitational Contraction. https://doi.org/10.1103/physrev.56.455
- R. et al. (2004). The Dynamics of General Relativity. https://doi.org/10.48550/arxiv.gr-qc/0405109
- Rines et al. (2013). MEASURING THE ULTIMATE HALO MASS OF GALAXY CLUSTERS: REDSHIFTS AND MASS PROFILES FROM THE HECTOSPEC CLUSTER SURVEY (HeCS). https://doi.org/10.1088/0004-637x/767/1/15
- Sanders (1999). The Virial Discrepancy in Clusters of Galaxies in the Context of Modified Newtonian Dynamics. https://doi.org/10.1086/311865
- Sanders (2003). Clusters of galaxies with modified Newtonian dynamics. https://doi.org/10.1046/j.1365-8711.2003.06596.x
- Sanders (2011). NGC 2419 does not challenge modified Newtonian dynamics. https://doi.org/10.1111/j.1745-3933.2011.01160.x
- Sartoris et al. (2020). CLASH-VLT: a full dynamical reconstruction of the mass profile of Abell S1063 from 1 kpc out to the virial radius. https://doi.org/10.1051/0004-6361/202037521
- Shao et al. (2012). New tests of local Lorentz invariance of gravity with small-eccentricity binary pulsars. https://doi.org/10.1088/0264-9381/29/21/215018
- Shin et al. (2019). Measurement of the splashback feature around SZ-selected Galaxy clusters with DES, SPT, and ACT. https://doi.org/10.1093/mnras/stz1434
- Shin et al. (2021). The mass and galaxy distribution around SZ-selected clusters. https://doi.org/10.1093/mnras/stab2505
- Skordis et al. (2021). New Relativistic Theory for Modified Newtonian Dynamics. https://doi.org/10.1103/physrevlett.127.161302
- Strohmayer et al. (2006). The 2004 Hyperflare from SGR 1806−20: Further Evidence for Global Torsional Vibrations. https://doi.org/10.1086/508703
- Tian et al. (2020). The Radial Acceleration Relation in CLASH Galaxy Clusters. https://doi.org/10.3847/1538-

4357/ab8e3d
- Tinto et al. (2014). Time-Delay Interferometry. https://doi.org/10.12942/lrr-2014-6
- Umetsu et al. (2016). CLASH: JOINT ANALYSIS OF STRONG-LENSING, WEAK-LENSING SHEAR, AND MAGNIFICATION DATA FOR 20 GALAXY CLUSTERS*. https://doi.org/10.3847/0004-637x/821/2/116
- Weisberg et al. (2016). RELATIVISTIC MEASUREMENTS FROM TIMING THE BINARY PULSAR PSR B1913+16. https://doi.org/10.3847/0004-637x/829/1/55